Positive Definiteness Failures in the pmsimstats Covariance Matrix: Prevalence, Causes, and Consequences

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Abstract

The `pmsimstats` data-generating process constructs a 26×26 covariance matrix from user-specified correlation parameters and passes it to `MASS::mvrnorm()` to draw simulated participant data. When the resulting matrix is not positive definite (PD), the code silently projects it to the nearest PD matrix via `corpcor::make.positive.definite()`. The simulation pipeline was instrumented to measure the frequency and severity of these failures across four code versions (commits). The results reveal that PD failure is not a rare edge case but a routine feature of the simulation: 24.7% of matrices fail at the initial commit, rising to 63.0% after the Ron Thomas edits. All 162 matrices in the parameter sweep are ill-conditioned (condition number > 100). The failures are concentrated in the OL+BDC design (83–92% failure rate) due to abrupt expectancy-driven variance changes mid-trial. These findings indicate that the correlation parameter space used in the publication is not self-consistent and that the PD correction introduces an uncontrolled perturbation to the intended covariance structure.

Architecture scope. The positive definiteness failures documented here are specific to Architecture B (MVN differential correlation), in which the biomarker-BR correlation parameter c_{bm} enters the off-diagonal entries of the covariance matrix. Under Architecture A (direct mean moderation), the BM-BR correlation block is absent from Σ and the PD constraint on c_{bm} is eliminated entirely: values up to 0.95 produce valid covariance matrices across all trial designs. For a comparison of both DGP architectures, see `02-dgp-mean-moderation-vs-mvn.tex`. For implementation details of Architecture A, see `19-mean-moderation-implementation-notes.tex`.

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1 Introduction

The `pmsimstats` package simulates clinical trial data by drawing participant outcomes from a multivariate normal (MVN) distribution. The MVN requires a positive definite covariance matrix Σ. The package constructs Σ from a correlation matrix R and a vector of standard deviations $\boldsymbol{\sigma}$:

$$\Sigma = \text{diag}(\boldsymbol{\sigma}) \cdot R \cdot \text{diag}(\boldsymbol{\sigma})$$

If the resulting Σ is not positive definite, the code applies a correction:

```
if (makePositiveDefinite) {
  if (!is.positive.definite(sigma)) {
    sigma <- make.positive.definite(sigma, tol = 1e-3)
  }
}
```

This correction projects Σ to the nearest PD matrix by setting negative eigenvalues to a small positive tolerance. The correction is silent: no warning is issued, no diagnostic is recorded, and the user has no way to know how many matrices were corrected or how severely the intended covariance structure was perturbed.

This white paper quantifies the prevalence and severity of PD failures across the parameter space used in the published Figure 4, traces their root causes, and evaluates the consequences for the simulation results.

2 Experimental Design

2.1 Instrumentation

The simulation was instrumented by replacing `corpcor::is.positive.definite()` and `corpcor::make.positive` with wrapper functions that record every call and its result. The wrappers additionally compute the eigenvalue spectrum and condition number ($\kappa = \lambda_{\max}/\lambda_{\min}$) for each matrix checked. No simulation code was modified; the instrumentation operates at the function-dispatch level.

2.2 Simulation Parameters

The parameter sweep is identical to the one used for Figure 4 in Hendrickson et al. (2020):

- **Trial designs:** OL (1 path), OL+BDC (2 paths), CO (2 paths), N-of-1 (4 paths)
- **Model parameters:** $N \in \{35, 70\}$, $c_{bm} \in \{0, 0.3, 0.6\}$, $t_{1/2} \in \{0, 0.1, 0.2\}$ weeks
- **Fixed correlations:** $c_{tv} = c_{pb} = c_{br} = 0.8$, $c_{cf1t} = 0.2$, $c_{cfct} = 0.1$
- **Censoring:** 4 patterns plus no censoring

2.3 Number of PD Checks per Simulation

The function `generateSimulatedResults()` calls `generateData()` once per path per parameter combination. Each call constructs one Σ matrix and checks it once for positive definiteness. The total number of checks is deterministic and independent of the number of replicates (`Nreps`), because all $N \times$ Nreps participants for a given path are drawn in a single `mvrnorm()` call:

Table 1: PD checks per simulation run

Design	Paths	× 18 param sets
OL	1	18
OL+BDC	2	36
CO	2	36
N-of-1	4	72
Total		**162**

2.4 Code Versions Tested

Four commits spanning the repository history were tested:

Table 2: Commits tested for PD failure rates

Commit	Description	Key change
`42ac030`	Initial (publication)	Step-function BM–BR correlation
`8609f12`	Ron Thomas edits	Proportional BM–BR scaling, $s = 2$
`f6ee86d`	Autocorrelation typo fix	Symmetric cross-factor correlation
`f70f86d`	Carryover term added	Additional DGP carryover logic

3 Results

3.1 Overall Failure Rates

Table 3: PD failure rates by commit (162 matrices per run)

Commit	Description	Failures	Rate (%)	Δ from initial
`42ac030`	Initial commit	40	24.7	–
`8609f12`	Ron Thomas edits	102	63.0	+38.3 pp
`f6ee86d`	Autocorr typo fix	102	63.0	+38.3 pp
`f70f86d`	Carryover added	102	63.0	+38.3 pp

The Ron Thomas edits (`8609f12`) more than doubled the PD failure rate from 24.7% to 63.0%. Subsequent commits did not change the rate, confirming that the additional failures are caused by the correlation scaling logic introduced at `8609f12`, not by the autocorrelation fix or carryover additions.

3.2 Condition Number Statistics

Every matrix in the parameter sweep is ill-conditioned. The condition number $\kappa = \lambda_{\max}/\lambda_{\min}$ measures numerical stability; values above 100 indicate ill-conditioning, and values above 10^6 indicate severe ill-conditioning.

At the initial commit, 75% of matrices have κ between 100 and 1,000 (ill-conditioned but not catastrophic). The remaining 25% (the failing matrices) have $\kappa > 10^{18}$, indicating that the negative

Table 4: Condition number statistics across all 162 matrices

Commit	Median κ	$\kappa > 100$	$\kappa > 1{,}000$	$\kappa > 10^6$
`42ac030`	229	162/162	40/162	40/162
`8609f12`	3.1×10^{18}	162/162	102/162	102/162
`f6ee86d`	3.1×10^{18}	162/162	102/162	102/162
`f70f86d`	3.1×10^{18}	162/162	102/162	102/162

eigenvalue is very small relative to the dominant eigenvalue. After the Ron Thomas edits, the median κ itself is 10^{18}, reflecting the majority of matrices now failing PD.

3.3 Eigenvalue Analysis of Failed Matrices

All PD failures involve exactly one negative eigenvalue. The severity ranges from barely negative (-0.52) to substantially negative (-33.3):

Table 5: Minimum eigenvalue in failed matrices

Commit	Worst	Median	Least severe
`42ac030`	-33.26	-15.36	-0.52
`8609f12`	-33.37	-17.11	-0.52
`f6ee86d`	-33.37	-17.11	-0.52
`f70f86d`	-33.37	-17.11	-0.52

A minimum eigenvalue of -33 is not a minor numerical perturbation. For context, the dominant eigenvalue is approximately 700 (the sum of the largest variances in the system). A negative eigenvalue of -33 represents approximately 5% of the dominant eigenvalue, indicating a substantial structural inconsistency in the specified correlation matrix.

3.4 Failure Breakdown by Design and Parameter

The failures are not distributed uniformly across designs. At the initial commit:

Table 6: PD failure rate by design and c_{bm} (initial commit). Each cell shows failures out of total matrices for that design-c_{bm} combination across all N and carryover values.

Design	$c_{bm} = 0$	$c_{bm} = 0.3$	$c_{bm} = 0.6$
OL	0/6 (0%)	0/6 (0%)	0/6 (0%)
CO	0/12 (0%)	1/12 (8%)	1/12 (8%)
N-of-1	2/24 (8%)	2/24 (8%)	2/24 (8%)
OL+BDC	**10/12 (83%)**	**11/12 (92%)**	**11/12 (92%)**

The OL+BDC design accounts for 32 of the 40 failures (80%). The OL design never fails. CO and N-of-1 have low failure rates that are largely independent of c_{bm}, indicating that the biomarker correlation is not the primary driver of PD failure for these designs.

4 Root Cause Analysis

4.1 Why OL+BDC Fails

The OL+BDC design has a distinctive feature that none of the other designs share: the expectancy value changes abruptly from 1.0 (open-label phase, first 4 timepoints) to 0.5 (blinded discontinuation phase, last 4 timepoints).

The expectancy value directly scales the standard deviation of the ER (expectancy response) factor:

$$\sigma_{\text{ER},t} = \sigma_{\text{ER}}^{\text{base}} \times e_t$$

where e_t is the expectancy at timepoint t. For OL+BDC:

$$\sigma_{\text{ER}} = [10, 10, 10, 10, 5, 5, 5, 5]$$

This creates a covariance matrix where the ER block has variance 100 for the first 4 timepoints and 25 for the last 4, while the cross-timepoint covariances (driven by the autocorrelation $c_{pb} = 0.8$) are:

$$\text{Cov}(\text{ER}_i, \text{ER}_j) = 0.8 \times \sigma_{\text{ER},i} \times \sigma_{\text{ER},j}$$

When i is in the open-label phase and j is in the blinded phase:

$$\text{Cov}(\text{ER}_i, \text{ER}_j) = 0.8 \times 10 \times 5 = 40$$

The corresponding correlation in the full covariance matrix is:

$$\text{Cor}(\text{ER}_i, \text{ER}_j) = \frac{40}{\sqrt{100 \times 25}} = 0.8$$

This is consistent. However, the problem arises from the interaction between the ER block, the other factor blocks (TV, BR), and the cross-factor correlations. The compound symmetry autocorrelation ($\rho = 0.8$) combined with cross-factor correlations ($c_{cf1t} = 0.2$, $c_{cfct} = 0.1$) and the heterogeneous ER variances creates a joint covariance structure that is not guaranteed to be PD.

Intuitively, the abrupt halving of ER variance mid-trial, while maintaining the same autocorrelation and cross-factor correlations, forces the correlation matrix into a region where the implied joint correlations are self-contradictory.

4.2 Why OL Never Fails

The OL design has constant expectancy ($e_t = 1$ for all t), producing homogeneous variances within each factor block. The compound symmetry structure with constant variances is PD by construction for the autocorrelation and cross-factor correlation values used (0.8, 0.2, 0.1). The BM–BR correlations (c_{bm} up to 0.6) do not violate PD in this homogeneous setting.

4.3 Why Ron Thomas Edits Increase Failures

The Ron Thomas edits introduced two changes that affect the correlation matrix:

1. **Timepoint-1 zero correlation.** The BM–BR correlation at the first timepoint is always set to 0, regardless of treatment status. This creates an additional asymmetry in the correlation matrix (the first BR column/row has zero correlation with BM while subsequent on-drug timepoints have c_{bm}).
2. **Proportional carryover scaling.** Off-drug timepoints with carryover residual receive a BM–BR correlation proportional to the residual ratio. At the carryover half-lives tested (0.1 and 0.2 weeks), these correlations are effectively zero (ratios < 0.001), creating additional near-zero entries in the correlation matrix that interact with the existing compound symmetry structure.

Both changes increase the heterogeneity of the BM–BR correlation row/column. At the initial commit, the BM–BR correlations are either 0 or c_{bm} (a simple binary pattern). After the edits, they can be 0, a tiny fraction of c_{bm}, or full c_{bm}, depending on the timepoint. This additional heterogeneity pushes more matrices below the PD boundary.

5 Consequences

5.1 What Does `make.positive.definite()` Do?

The function `corpcor::make.positive.definite(m, tol = 1e-3)` performs the following:

1. Computes the eigendecomposition: $\Sigma = Q\Lambda Q^T$
2. Replaces any $\lambda_i <$ tol with tol
3. Reconstructs: $\Sigma^* = Q\Lambda^* Q^T$

This is the "nearest PD matrix" in the Frobenius norm sense. However, it does not preserve the structure of the intended correlation matrix. Specifically:

- The diagonal elements of Σ^* may differ from the intended variances.
- The off-diagonal elements (covariances) are perturbed by an amount proportional to the magnitude of the negative eigenvalue.
- The implied correlation matrix $R^* = D^{-1}\Sigma^* D^{-1}$ (where $D = \text{diag}(\sqrt{\text{diag}(\Sigma^*)})$) may have entries that differ substantially from the intended values.

5.2 Magnitude of the Perturbation

For the worst-case failures (minimum eigenvalue ≈ -33), the correction shifts eigenvalues by up to 33 units. Given that the total variance in the system (trace of Σ) is approximately $\sum_i \sigma_i^2 \approx 3{,}200$, the perturbation represents roughly 1% of the total variance. However, because the perturbation is concentrated in specific eigenvectors (those associated with the near-zero or negative eigenvalues), it can disproportionately affect particular correlation entries (especially the BM–BR correlations and cross-phase ER correlations that created the non-PD condition in the first place).

5.3 Impact on Power Estimates

The practical impact on power estimates is difficult to quantify analytically because the perturbation is parameter-dependent and interacts with the analysis model in complex ways. However, several qualitative observations can be made:

1. **The OL+BDC design is most affected.** With 83–92% of its matrices requiring correction, the OL+BDC power estimates are based almost entirely on perturbed covariance matrices. The intended correlation structure is not faithfully represented.
2. **The OL design is unaffected.** No corrections are needed, so the OL power estimates reflect the intended covariance structure exactly.
3. **Design comparisons are confounded.** When comparing power across designs (the paper's primary analysis), the OL+BDC results are based on a different effective covariance structure than the OL and CO results. This is a systematic confound that cannot be resolved by increasing the number of replicates.
4. **The Ron Thomas edits worsen the confound.** By increasing the failure rate from 25% to 63%, the edits ensure that the majority of all simulation runs across all designs operate on perturbed covariance matrices.

6 Relationship to the Correlation Scaling Artifact

This PD failure analysis is related to but distinct from the correlation scaling artifact described in the companion white paper. The two issues are:

1. **Correlation scaling artifact:** The original code assigns full BM–BR correlation to off-drug timepoints with negligible carryover residual, creating spurious conditional expectations. This is a *modeling choice* issue.
2. **PD failure:** The specified correlation parameters do not produce a valid covariance matrix for certain design-parameter combinations, requiring silent correction. This is a *parameter validity* issue.

The two issues interact: the correlation scaling rule determines which BM–BR correlation values are assigned, which in turn determines whether the resulting matrix is PD. The Ron Thomas proportional scaling creates more heterogeneous BM–BR correlation patterns, which increases PD failures. However, even under the original (simpler) correlation rule, the OL+BDC design fails PD over 80% of the time due to the expectancy-driven variance heterogeneity.

7 Recommendations

7.1 Short-Term Mitigations

1. **Log PD failures.** Add a counter to `generateData()` that records how many matrices required PD correction, and include this count in the simulation output. This allows users to assess the prevalence of corrections for any parameter configuration.
2. **Warn on high failure rates.** If the PD failure rate exceeds a threshold (e.g., 10%), issue a warning indicating that the correlation parameters may not be self-consistent for the specified design.
3. **Report the perturbation magnitude.** When a correction is applied, compute and log the Frobenius norm of the perturbation: $\|\Sigma^* - \Sigma\|_F$. This allows users to assess whether the correction is negligible or substantial.

7.2 Medium-Term Solutions

1. **Pre-validate parameter combinations.** Before running the simulation, test each unique (design, model parameter) combination for PD and reject or flag invalid ones. The `pmsimstats2025` repository implements this via `validate_parameter_grid()`.

2. **Constrain the correlation parameter space.** Determine the feasible region of $\{c_{tv}, c_{pb}, c_{br}, c_{cf1t}, c_{cfct}, c_{bm}\}$ that guarantees PD for each design. This can be done numerically by sweeping the parameter space and testing PD at each point.

3. **Address the OL+BDC variance heterogeneity.** The OL+BDC design's 83% failure rate is driven by the abrupt expectancy change. Consider whether the compound symmetry autocorrelation assumption is appropriate when variances change mid-trial, or whether a different autocorrelation structure (e.g., damped exponential) would be more appropriate.

7.3 Long-Term Structural Solutions

1. **Cholesky parameterization.** Instead of specifying a correlation matrix and checking PD after the fact, parameterize the covariance matrix via its Cholesky factor L such that $\Sigma = LL^T$ is PD by construction. This eliminates the need for PD correction entirely but requires reparameterizing the correlation structure.

2. **Structured correlation models.** Replace the ad hoc compound symmetry + BM–BR correlation with a structured model (e.g., factor-analytic, spatial exponential) that has known PD constraints. This would make the feasible parameter region explicit rather than requiring numerical validation.

3. **Separate the correlation parameters from the variance parameters.** The current approach specifies correlations and variances independently, then combines them into a covariance matrix. Because PD depends on the joint structure, parameters that are individually reasonable can produce an invalid joint specification. A parameterization that enforces joint validity would prevent this class of errors.

8 Conclusions

1. Positive definiteness failure is not a rare edge case in the `pmsimstats` simulation pipeline. At the parameter values used in the publication, 24.7% of covariance matrices require silent correction at the initial commit, rising to 63.0% after the Ron Thomas edits.

2. All 162 matrices in the parameter sweep are ill-conditioned ($\kappa > 100$), and the failed matrices have condition numbers exceeding 10^{18}. The negative eigenvalues are substantial (up to -33), not mere floating-point artifacts.

3. The failures are concentrated in the OL+BDC design (83–92% failure rate) due to the abrupt expectancy-driven change in ER variance mid-trial. The OL design never fails.

4. The `make.positive.definite()` correction silently perturbs the intended covariance structure. Because the perturbation is design-dependent, cross-design power comparisons (the paper's primary analysis) are confounded by differences in the effective covariance structure.

5. The Ron Thomas correlation scaling edits increase the PD failure rate by introducing additional heterogeneity in the BM–BR correlation pattern. This compounds the existing OL+BDC

failures with new failures across all designs.

6. The correlation parameters used in the publication ($c_{tv} = c_{pb} = c_{br} = 0.8$, $c_{cf1t} = 0.2$, $c_{cfct} = 0.1$, c_{bm} up to 0.6) are not self-consistent for the OL+BDC design. Any simulation results for this design should be interpreted with the understanding that the covariance structure was machine-corrected rather than as specified.